



**Sophia C.**

ID: QNVO

**Gender:** Female

**Age:** 72

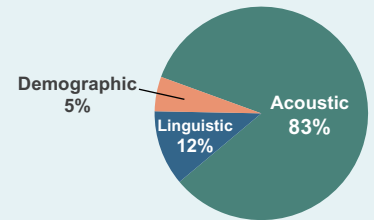
**Education:** Technical/Undergraduate

**Primary Language:** English

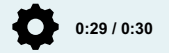
**Probability of Predicted Outcome: 80.1%**

- ☐ Healthy
- ☒ MCI
- ☐ AD

**Contribution of each modality to decision making**



**Listen to the voice!**



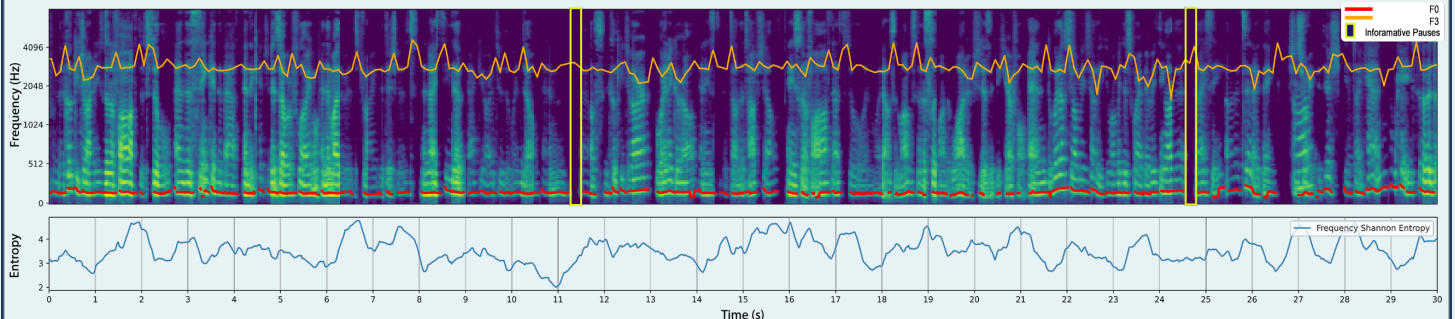
Transcription

from the cookie jar and he's going to fall off the stool and the sink in the bath and the kitchen is overflowing and the mother is drying the dishes and I see the backyard through the window and what else the cookie jar boy and a girl and he's it's on the top shelf and he's going to fall off that stool and what else the mom's going to slip on the water she doesn't be careful and what else you want me to tell you do you want me to describe everything on that that I see oh that's it I think she's drying the dish yes I can okay so the

Interpretation

- Frequent repetition of specific words or phrases (e.g., “me,” “what,” “I see”) may indicate difficulty with lexical retrieval, suggesting memory challenges.
- Disfluencies and filler phrases (e.g., “what else”) highlight issues with maintaining fluent speech, pointing to possible disruptions in thought organization.
- Persistent first-person focus (e.g., “I” statements) reflects limited variation in language use and a reduced ability to shift perspectives.
- Reduced narrative coherence (e.g., abrupt topic shifts or minimal detail) indicates challenges in planning and structuring discourse, a sign consistent with mild cognitive impairment.

Spectrogram + SHAP Visualization



Interpretation

- A few instances of informative pauses (e.g., pauses before nouns) were detected.
- Minimal signs of low energy levels were observed in the voice (see Spectrogram highlighted in green).
- Speech was relatively rhythmic, with minimal flat or monotonous segments (see Entropy curve).
- The voice was almost stable (standard deviation of Shimmer was relatively high).
- Control over vocal folds was acceptable. (see F0 curve).
- Tongue–lip coordination for phoneme and syllable production was acceptable (see F3 curve).

Overall, there were some signs of cognitive impairment, suggesting the individual has likely mild cognitive impairment.

## CONSIDERATIONS:

**Please be advised that the sensitivity of this system is not 100%. A more comprehensive evaluation should include the individual's medical history and additional cognitive assessments.**

To reduce the risk of cognitive status, evidence-based studies suggested:

- Having regular exercise
- Connecting with family/community
- Limiting alcohol intake

